

HC HANDBOOK

#scienceoflearning

Evaluate and use effective strategies to learn or teach specific types of material.

Please fill out this [Google Form](#) to submit suggestions for this page (i.e. typos, study guide contributions, useful applications, etc.). We are actively seeking feedback on anything big or small to make the HC Handbook as useful as possible.

Defining the HC

Pitch

Learn how to learn.

Paragraph Description

All of us are lifelong learners. We do not learn only in classes, but instead continually must learn new material to adapt to changing circumstances and to expand our horizons. Some learning techniques (e.g., testing yourself) have proven very effective for many purposes whereas others (e.g., underlining material in a book or article) have proven effective only for a narrow range of purposes. To know when to use a given technique, you must know why the techniques are effective for specific purposes—which relies on understanding principles from the science of learning. These principles are effective both for learning on your own and teaching others.

Categorization

Habit of Mind | Empirical Analyses | Thinking Creatively | Solving Problems

Cornerstone Introduction



Class: Empirical Analyses | Semester One

Unit: Problem Solving

Big Question: How can we feed the world?

General Example

You want to learn a foreign language. You use practice testing to learn vocabulary, knowing that it is successful in part because it elicits the generation effect; this effect states that information is more likely to be remembered if it is generated by one's own mind as opposed to being passively read from a source. You realize that you can improve your use of the practice testing learning technique by considering the learning principle of dual codes in addition to the learning principle of generation effect. Dual codes states that you are more likely to recall the same material if processed in two different media, because it enriches their retrieval cues. To incorporate dual codes into your use of practice testing, you create flashcards of pictures, to which you will answer the vocabulary word the picture represents. In addition, you space your practice out with increasing gaps, answering those vocabulary questions correctly multiple times to ensure you can easily use those words in conversation when you travel.

Footnote: *An appropriate learning technique has been used (practice testing) and implemented in an optimal way by considering how the technique can draw on additional learning principles. In addition, the schedule of the study sessions is addressed in such a way that will increase retention.*

Is it this HC?

Is #scienceoflearning the right HC? If the answer to the following questions is yes, #scienceoflearning could be useful:

1. Are you considering how study techniques, shortcuts, or methodologies were implemented to learn or remember something?
2. Did you evaluate a particular learning technique or learning principle?

Depending on the work product, there might be other HCs to consider. Not all analyses related to learning are #scienceoflearning.

The following HC might apply if the work:

#selfawareness

- generates a self-assessment of one's study performance with context
- explains the factors that inhibit or enhance learning progress

#psychologicalexplanation

- uses psychological theories to explain learning behavior

#biasidentification

- describes the potential cognitive biases in certain study techniques (e.g. attentional bias in highlighting).
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Self Study

Try answering the questions on your own before checking the example answers.

 **Identify the seven common learning techniques from Dosmann & Stein (2019), with a brief description of how they work.**

Answer:

- **No strategy:** Unsurprisingly, this is the least effective approach to studying, but it is commonly used. Students too often just read assigned material without applying any strategy.
- **Self-explanation:** The technique of self-explanation involves stopping while reading assigned material or solving a problem and giving a short explanation of a portion of that material to yourself in your own words.

- **Practice testing:** While you read assigned material, or after, come up with problems or short questions (with answers) about the material that you can ask yourself later.
- **Re-reading:** This simply involves reading assigned material for a second, third, and fourth time (and beyond). The repeated readings could happen immediately after previous readings, or at different time lags, such as after one day or one week.
- **Highlighting/underlining:** Highlighting involves marking portions of the text you read, such as vocabulary words, sometimes to review later.
- **Summarization:** Summarization involves taking sections of an assigned reading, such as the conclusion of an article, and writing a short description about the main points.
- **Spaced practice:** Whatever strategy you use, if you repeat it after a time lag, you are engaging in spaced practice.

👉 **What are the two maxims according to Kosslyn (2017)?**

Answer:

Maxim 1: Think it Through (The more you think something through, paying attention to what you are doing, the more likely you are later to remember it.)

Maxim 2: Make and Use Associations (Associations not only help us organize material so that it is easy to store in memory but also give us the hooks that will allow us later to retrieve the material from memory.)

👉 **What are the 16 learning principles according to Kosslyn (2017)?**

Answer:

Maxim 1: Think it Through (6 principles)

Focus: More processing of the relevant information will produce better memory

- Deep processing
- Desirable difficulty
- Generation effect

Focus: Ways to induce additional information processing

- Interleaving
- Dual codes
- Emotion

Maxim 2: Make and Use Associations (10 principles)

Focus: Structure information by using associations

- Chunking
- Build on prior associations
- Foundational learning
- Deliberate practice
- Appropriate examples
- Principles, not rote

Focus: Create rich retrieval cues

- Associative chaining (story telling)
- Spaced practice
- Different contexts
- Avoiding interference

👉 How can rereading improve learning?

Answer: Rereading is useful when you need conceptual organization of the new material and distinction between important concepts. However, the technique has the lowest yield when rereading happens immediately. This is because humans may focus on the same parts as during the first read. Rereading, although generally less effective, can still enhance learning if it

happens after time delay and with a specific goal in mind. What are you trying to get out of the re-read?

👉 How can you enhance the note-taking technique using learning principles?

Answer: Some examples:

- **Deep processing:** Reading texts, extracting useful information, organizing information, and writing down important notes all require a lot of mental processing, which is further enhanced when using different sources
- **Dual codes:** Add visualizations to synthesize different pieces of information. For example, drawing a V-shaped water molecule to understand its polarity.
- **Chunking:** Group information into chunks (organized units). For example, organizing HCs based on the cornerstone courses and/or core competencies.
- **Appropriate examples:** Associate abstract concepts using real-life examples. For example, applying probability concepts in a gambling example.

Note 1: There is no specific answer to this question! Every one has a different note-taking approach, which often utilizes combinations of learning principles. The list above is not exhaustive, and you're not expected to use all the learning principles.

Note 2: Note-taking is less effective because it is often used to remove the mental load by storing the information externally rather than in one's memory. However, it can enhance learning when used appropriately. Since most of its usefulness comes from the process itself, reviewing another student's notes is often ineffective.

💡 Applying the HC

Guided Reflection

- Can you demonstrate understanding of the selected learning principles?

- Have you reflected on how a particular learning technique improved/impaired your learning?
- Have you explained how underlying learning principles support a study technique?
- Have you provided evidence to demonstrate the use of different learning techniques and/or principles?
- Have you explained why a particular learning technique or learning principle is beneficial for the given context or material being studied?
- Have you modified a study technique to tailor it to a particular learning task, justifying the modification?
- Have you explained why certain combinations of principles/study techniques are complementary to each other?
- Have you critiqued a certain study technique for its strengths and weaknesses?

Common Pitfalls

- The application confuses study techniques and learning principles.
- The application classifies a study technique as being effective but does not support their arguments with any learning principles.
- The application lists learning principles underlying a study technique without any explanation.
- The application describes a learning principle but the principle is not related to the study technique of interest.
- The application misclassifies the effectiveness of a particular learning technique for a particular context.
- The application is incorrect or unclear in how the study technique works.

Applying the HC at Minerva

- You can use this HC in any of the courses you take at Minerva. You will identify techniques that work for you, or even design one by yourself and modify it based on the content of the course. For example, you might find practice testing more effective in FA50 than MC50 because you often have to solve textbook problems as part of pre-class work.
- In tutorial classes, students draw on the learning principles when designing Forum sessions. Some tutorials treat this HC as an LO because it is so often used.

- When going to a new rotation city, you might want to learn the language. Combining different learning techniques may help you learn the language efficiently.
- In the “Self Study” section, the toggle switches act like flashcards, nudging you to think about the questions before revealing the answers (generation effect)!

Applying the HC Outside of Minerva

- Mastering this HC is essential for every student regardless of their field. As lifelong learners, one needs to know the learning techniques that have been empirically tested as well as discover what works for them.
- This HC is especially useful if you see yourself contributing to the design of future educational institutions (like Minerva!). The knowledge of learning principles is useful in designing curricula.

Key Resources

- The science of learning. In S. M. Kosslyn & B. Nelson (Eds.), Building the Intentional University: Minerva and the Future of Higher Education. https://course-resources.minerva.edu/uploaded_files/mke/00069173-9345/kosslyn--s.--2017--the-science-of-learning.pdf
 - This review provides an overview of the principles of human learning. Focus on describing how the principles aid learning.
- Minerva University (2019). Common learning techniques: Using #scienceoflearning to make the most of your study time. <https://my.minerva.edu/academics/hc-resources/cornerstone-custom-hc-guides/>
 - This resource describes common learning techniques: methods you use to acquire new skills and knowledge. Refer to the citations of this reading if you want to learn more about #scienceoflearning.
- Etoile Boots Finley Draffin-Jacquín, Zhi Zhi Chia (2022). Approaching Dense Readings. <https://my.minerva.edu/academics/learning-resources/science-learning/approaching-dense-readings/>

- This whitepaper provides general guidelines for approaching different types of readings you will come across at Minerva such as, scientific articles, computational texts and readings in humanities.
- Etoile Boots (2022). Approaching the 90-Minute Class.
[Interactive Infographic]
<https://my.minerva.edu/academics/learning-resources/science-learning/approaching-90-minute-class/>
 - This interactive infographic explains the reasoning behind Minerva class structure and provides tips on making the most of the in-class learning experience.